

Python Code

```
1  """
2  Real-Time Cyberbullying Detection – Inference Demo
3  Run AFTER main.py has been executed.
4  """
5  import warnings
6  warnings.filterwarnings('ignore')
7
8  import sys, os
9  sys.path.insert(0, '.')
10
11  from cb_main import TextPreprocessor, CyberbullyingDataGenerator
12  import numpy as np
13  import joblib
14  import time
15
16  def load_models():
17      print("[Loading models...]")
18      models = {
19          'Naive Bayes':      joblib.load('models/naive_bayes.pkl'),
20          'Logistic Regression': joblib.load('models/logistic_regression.pkl'),
21          'Linear SVM':      joblib.load('models/linear_svm.pkl'),
22          'Random Forest':   joblib.load('models/random_forest.pkl'),
23      }
24      vectorizer = joblib.load('models/tfidf_vectorizer.pkl')
25      le         = joblib.load('models/label_encoder.pkl')
26      prep       = joblib.load('models/text_preprocessor.pkl')
27      print("All models loaded!\n")
28      return models, vectorizer, le, prep
29
30
31  def predict_single(text, models, vectorizer, le, prep):
32      cleaned = prep.clean(text)
33      X       = vectorizer.transform([cleaned])
34      normal_idx = le.transform(['normal'])[0]
35
36      votes = {}
37      for name, model in models.items():
38          pred = model.predict(X)[0]
39          votes[name] = {
40              'label':    le.inverse_transform([pred])[0],
41              'is_bully': int(pred != normal_idx)
42          }
43
44      bully_count = sum(v['is_bully'] for v in votes.values())
45      ensemble    = bully_count >= 2
46      confidence   = bully_count / len(models) * 100
47
48      return votes, ensemble, confidence
49
50
```

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51 def run_demo():
52     if not os.path.exists('models/naive_bayes.pkl'):
53         print("Models not found. Please run main.py first.")
54         sys.exit(1)
55
56     models, vectorizer, le, prep = load_models()
57
58     # Test comments – mix of normal and cyberbullying
59     test_comments = [
60         "This is such a great video, I really enjoyed watching it!",
61         "You are so stupid, nobody wants to hear your opinion.",
62         "Thanks for sharing this helpful information with us.",
63         "I hope something terrible happens to you for posting this.",
64         "Great explanation, very easy to understand the concept.",
65         "You are the dumbest person I have ever encountered online.",
66         "This made my day, absolutely wonderful content keep it up.",
67         "People like you should not be allowed on the internet.",
68         "I disagree with your point but I respect your opinion.",
69         "Say that again and see what happens to you.",
70         "Really appreciate the effort you put into making this.",
71         "What an idiot, completely brainless comment you made.",
72         "Such a positive community, glad to be part of this.",
73         "You deserve every bad thing that comes your way.",
74         "Looking forward to more content like this from you!",
75     ]
76
77     div = "=" * 72
78     print(div)
79     print(" REAL-TIME CYBERBULLYING DETECTION DEMO")
80     print(div)
81     print(f"{'#':>3} {'Prediction':<14} {'Conf':>6} {'Text Preview':<35}")
82     print("-" * 72)
83
84     correct = 0
85     true_labels = [0,1,0,1,0,1,0,1,0,1,0,1,0] # ground truth
86
87     for i, comment in enumerate(test_comments):
88         votes, ensemble, confidence = predict_single(
89             comment, models, vectorizer, le, prep)
90         pred_str = '[CYBERBULLYING]' if ensemble else '[ NORMAL ]'
91         preview = comment[:35] + '...' if len(comment) > 35 else comment
92
93         if ensemble == true_labels[i]:
94             correct += 1
95
96         print(f"{i+1:>3} {pred_str:<14} {confidence:>5.0f}% {preview}")
97
98         # Show model votes
99         vote_str = ' '.join([f"{n.split()[0]}:{v['label'][:5]}"
100                                for n, v in votes.items()])
101         print(f"      Votes: {vote_str}")
102         time.sleep(0.05)
103
104     print("-" * 72)

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105     print(f"\n Accuracy: {correct/len(test_comments)*100:.1f}%  "
106           f"| Correct: {correct}/{len(test_comments)}")
107     print(div)
108
109     # Interactive mode
110     print("\n[INTERACTIVE MODE] Type a comment to classify (or 'quit' to exit):")
111     print("-" * 72)
112     while True:
113         try:
114             user_input = input("Enter comment: ").strip()
115             if user_input.lower() in ['quit', 'exit', 'q', '']:
116                 print("Exiting demo.")
117                 break
118             votes, ensemble, confidence = predict_single(
119                 user_input, models, vectorizer, le, prep)
120             result = "CYBERBULLYING DETECTED" if ensemble else "Normal - No threat"
121             print(f" Result      : {result}")
122             print(f" Confidence : {confidence:.0f}%")
123             for name, v in votes.items():
124                 print(f" {name:<22}: {v['label']}")
125             print()
126         except (EOFError, KeyboardInterrupt):
127             break
128
129 if __name__ == '__main__':
130     run_demo()
131
```